

Xiemin (Simon) Wen

Email: x5wen@uwaterloo.ca Phone: +1 (226) 234-8126 Personal Website: xieminwen.com

EDUCATION

University of Waterloo

Waterloo, Ontario

Major: **Computer Science** (GPA: 3.7/4.0)

Sept 2023 - Present

PROFESSIONAL EXPERIENCE

QA Automation Intern – Tangam System (Software company for casino analytics)

May 2026 - Aug 2026

- Developed and maintained 200+ automated end-to-end test cases across 70+ product tickets using Cypress, JavaScript, and Cucumber for a casino analytics platform.
- Automated regression testing for user-facing web workflows, validating feature behavior, data presentation, and cross-page functionality through reusable E2E test scenarios.
- Designed and validated 10+ automation scenarios across four major product areas during a focused testing cycle, expanding regression coverage for high-priority product workflows.

QA Analyst Intern – Propel Holdings (AI-powered fintech company)

Sep 2025 - Dec 2025

- Performed SQL-driven backend validation across 100+ QA tickets, verifying customer and loan-application data across databases, APIs, UI output, and business-rule logic.
- Developed reusable SQL queries to reconcile backend records with API responses and user-facing data, identifying inconsistencies across financial product workflows.
- Validated data integrity and application behavior across database, backend, and frontend layers for production financial systems and lending workflows.

Project Lead - Microsoft & University of Waterloo

May 2024 - Aug 2024

- Led a 4-person engineering team in developing an AI-powered investment insights platform that generated personalized guidance from users' financial profiles, investment goals, and market interests.
- Co-developed the AI, backend, and database layers using Python, Microsoft Azure, and Azure SQL to support investment insights, suggested allocations, and financial-report analysis.
- Owned project planning, technical coordination, team meetings, and final delivery, earning an Outstanding overall performance rating for the work term.

PROJECTS

AI Artifact Store

July 2026 - Aug 2026

(C++20, Distributed Systems, PostgreSQL, Boost.Asio/Beast, SHA-256)

- Engineered a distributed content-addressed artifact storage system in C++20 using SHA-256 content addressing, FastCDC/fixed-size chunking, PostgreSQL metadata, and Boost.Asio/Beast networking.
- Implemented resumable push/pull transfers, multi-node replication and repair, garbage collection, lifecycle management, and integrity verification across storage nodes.
- Built 415 automated tests (409 GoogleTests + 6 process-level E2E tests) and benchmarked FastCDC at ~95.8% chunk reuse under shifted-content workloads versus 0% for fixed-size chunking.

Graph-Augmented Code Intelligence Engine

July 2026 - Present

(Python, Static Analysis, Graph Algorithms, Information Retrieval)

- Architected a code-intelligence engine that models repositories as semantic code units and structural relationships—including calls, imports, references, inheritance, and containment—to improve code retrieval.
- Building a Python-based static-analysis pipeline with pluggable language adapters for extracting semantic code structure and constructing graph-aware representations of source repositories.
- Designed a retrieval architecture combining lexical and dense retrieval, controlled graph expansion, and structured reranking to evaluate graph-augmented context selection against text-only baselines.

SUMMARY OF QUALIFICATIONS

Technical Skills: C++, Python, JavaScript, SQL, Java, C#, Distributed Systems, Backend/API Development